SI 650 / EECS 549 F26 Project Proposal

Due: See Canvas for official deadline

1 Introduction

The course project is intended to provide an opportunity for students to dive deeper into one problem or topic of their choice and write a very small scale study on the topic. Projects may be done individually or in small teams, as described in the syllabus. Ideally, your course project is a chance to develop something you can talk about during job interviews to show deep experience in a topic, code you can show off to future employers to demonstrate technical expertise, or the results could serve as a pilot study for a full research project. Following, we outline the key requirements for your proposal.

2 Project Proposal Deliverables

Introduction (10 points) You should have a rough draft of the introduction that states what the problem is and provides some broader context. You should also include a statement on why solving this problem matters. In short, who would care if you solved it and what effect would solving it have? We recommend writing the introduction after you finish the IR Task Definition and Related Works sections. You will have a clearer sense after those sections are written.

IR Task Definition (20 points) The section describes what specific problem or IR task you plan to solve and goes into much more detail than what's specified in the Introduction. Give example of potential queries and documents in the collection. If your system is query-less, what is the implicit information need? You should add details and examples about what your problem is. It helps to be specific about what kind of input your system will use and what specifically it will produce as output.

At this stage, you do not need to know how to solve your problem! We are asking you to think of an information need you could address with an IR system and then to figure out how to describe that as a specific task. Tasks have inputs and outputs, which you can use to explain your system. For example, does your system only take text as queries, or will it need other things (networks? images? twitter bios? biomedical data?) and what kinds of things will it retrieve (movies? documents? images? recipes?).

Data (20 points) The section describes what data you will use for the project. You need to be specific about what the queries are in your system. How will (or did) you collect it? How is relevance defined for your task?

If you have the data already (or can get it), you should describe the source of the data, how you obtained it, what type of preprocessing steps you took, and (if not sensitive data) include a few examples. We strongly encourage you to include some very rough statistics (e.g., how many instances you have, the class distribution if doing classification, relevance score distribution) in a table format.

If you don't have your data, you should go into detail about how you will acquire it. This may include web scraping, cleaning, requesting forms, etc. We do not recommend using data that will require getting permissions, as you may be waiting a while. You should mention a back-up data source if your current data source falls through.

For all data, describe where your relevance scores will come from, which will be needed for evaluating. Many students have to collect these by annotating themselves, which is fine.

Related Work (10 points) The related work section should describe how other people have thought about the problem you're working on. How did they approach it? What makes their problem different from yours? Why do you think your approach will be better? For your proposal, you should have at least five papers related to your current problem and a few sentences describing what they did to solve the problem. We recommend using Semantic Scholar or Google Scholar to help find related papers.

Evaluation and Results (30 points) You **must** include a specific evaluation metric as a part of the proposal. In essence, **you need to define how you will measure how good your IR system is at your particular task or at solving your problem.** Manual analysis of output can be helpful, but you need to know quantitatively how to measure goodness[1].

If you're not sure about a good evaluation, please post on Piazza and we can work it out there. We strongly recommend looking at the related work papers you find and considering how those papers evaluate their systems. You are encouraged to think about multiple evaluation metrics— what might matter to people who use your system?

The second part of the evaluation plan is what baselines you'll compare your system against. One baseline should be random performance—e.g., returning a random document from the collection. A second baseline should be something reasonable that doesn't require much knowledge or learning, like BM25 with no hyperparameter tuning. A baseline is essential here because it helps you figure out what is the simplest way to solve your task. Baselines provide a useful comparison for putting your model in context.

Work Plan (10 points) Describe your work plan for the semester in terms of (i) what you've done so far and (ii) what you intend to do to finish the project. Be specific and lay

[1]Projects involving topic modeling or clustering a document collection usually have a very hard time coming up with a precise metric (i.e., how good is one topic model versus another?), so we recommend avoiding those unless you have an extrinsic tasks that you can test on (i.e., topic modeling is used as a method to facilitate solving some other kind of IR task).

out weekly objectives/milestones that you can use to keep track of progress. We won't hold you to this workplan for the project update or final report, but we have found that the act of creating such a work plan generally helps students quantify the remaining work involved and scope their project better (fewer December surprises!). Your work plan needs to be detailed and concrete and follow the SMART goals guidelines.

3 What to submit?

Please submit:

1. A PDF of your proposal
2. Your prompting and chat history related to *any* genAI use while working on this assignment. **Note that this is a required component even if you didn't use genAI. If that is the case, simply include a document that says you did not use genAI.**

4 What to expect next?

The first project update (Week 11) serves two purposes. The first is that it helps you write an initial rough draft of your final project report. The second is that it requires you to have all of your data in hand and to develop a simple baseline for your proposed task. Both of these are important steps at keeping your project on track.

The second update will occur during the lecture and discussion sections during Week 14. During lecture time, you will be paired with 1-2 other teams to talk to. During the discussion section, each team will present the status of their project and ask questions/feedback/help from other teams.

5 Late Policy

Throughout the semester, you have five free late days total. These are counted as whole days, so 1 minute past deadline results in 1 late day used (Canvas tracks this so it's easier/fair). However, if you have known issues (conference, etc.) let us know at least 24 hours in advance and we can work something out. If you are dealing with an extreme situation (family/health issue), let the instructional team know and we will figure out a path forward. Once the late days are used up, the homework cannot be submitted for a second time. Though speak with the instructor first if you think there's a legitimate reason it should be, rather than simply not submitting.

6 Academic Honesty Policy

Unless otherwise specified in an assignment, all submitted work must be your own work. Any excerpts, statements, or phrases from the work of others must be clearly identified as a quotation, and a proper citation provided.

GenAI use: It is ok to use genAI as a tool to help you ideate and to polish your proposal. But do not use it to directly generate the proposal for you. Note that you need to 1) not use genAI to generate the answer (or parts of the answer) but ask clarification and conceptual questions and 2) turn in the set of prompts and/or chat history.

If you are in doubt about any part of the policy, please contact one of the instructors to check.